

REAL-TIME ADAPTIVE LANE DETECTION FOR INDIAN ROADS

A Major Project report submitted
in partial fulfillment of requirement for the award of degree

BACHELOR OF TECHNOLOGY
in
SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE
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CERTIFICATE

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ABSTRACT

One of the most important technologies in advanced driver assistance systems and autonomous driving is lane detection, which enables a vehicle to go through the route and eliminate potential hazards. This project was designed to build a robust lane detection system for Indian roads because Indian roadways pose their own specific challenges such as faded or inconsistent lane markings, varied road conditions, diverse traffic patterns, and frequent obstructions. This system proposes to detect lane boundaries accurately and in real time using a combination of advanced image processing methods, computer vision tools and deep learning approaches such as OpenCV. Techniques like edge detection, Hough Transform and HSV color space for color based filtration are adapted to improve performance in diverse conditions. The project aims to optimize it for practical use on Indian roads, so it should be able to handle missing lane markings and irregular road layouts. Preliminary results show that considerable improvements in detection accuracy and robustness over conventional approaches are obtained. This project addresses the immediate need for safer and more reliable navigation tools but contributes, at large, to the adoption of ADAS and autonomous driving technologies in India, which can be a critical foundation for future intelligent transportation systems tailored for emerging markets.

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CHAPTER-1

INTRODUCTION

Lane detection is the basic step in navigation, lane departure warning, and path planning for advanced driver assistance and self-governing vehicles. While there are many developments globally towards lane detection technologies, the systems still face challenges to adapt to the Indian road conditions as they are unique and more diverse. Indian roads have faded lane markings with worn-out or non-existent lane signs, unpredictable behavior of the traffic, and road blockages such as parked vehicles, pedestrians, and animals. Environmental conditions are an added source of difficulty, with different lighting conditions, seasonal rains, and road deterioration that interfere with reliable lane detection.

This project develops a lane detection system for Indian roads by overcoming all these difficulties. The system utilizes state-of-the-art computer vision for real-time identification of lane boundaries, even under difficult conditions. Images go through preprocessing, which removes the noise, detects features using image segmentation techniques and edge detection techniques for accurate lane identification.

The significance of the project lies in overcoming most of the difficulties faced in lane detection on Indian roads, it adds up to the greater goal of developing intelligent transportation systems for raising markets as it creates a safer and more efficient drive.

1.1 EXISTING SYSTEM:

Although many methods like edge detection and the Hough Transform are available for the traditional techniques of image processing, their performance decreases when the complexity in road conditions becomes very serious. Approaches through machine learning, consisting of algorithms like SVM, provide a degree of adaptability and flexibility. However, they are still limited concerning the manipulation of the address of numerous scenarios in real-world applications. Applications with deep learning-based approaches, including advanced U-Net and LaneNet, show significantly higher accuracies compared to conventional approaches. However, deep learning methods generally need large sets of training data and incur high computational costs to achieve their respective gains. Hybrid systems were proposed concentrating on combining the gains both in terms of efficiency as well as accuracy by providing the benefits of traditional approaches along

with the advanced deep learning methods. These hybrid systems are known to work well on structured roads in commercial ADAS applications. However, they tend to face significant challenges and hardship when applied to Indian roads, characteristically less structured and more complex in nature. All these approaches, however, have crucial shortcomings in problems related to missing lane markings, high computational demand by advanced algorithms, and issues of adaptability by unstructured driving conditions.

1.2 PROPOSED SYSTEM:

The proposed lane detection system is particularly designed to address the challenging conditions commonly seen on Indian roads. These include inconsistent or faded lane markings which cause significant impairment to visibility and cluttered views from various vehicles, pedestrians, and the unpredictable changes in light and weather. Under these conditions, lane detection is a bit more complicated, requiring a highly flexible system that would adapt to these diverse conditions yet remain accurate in its detections.

Key Features

- **Straight Lane Detection:**

By applying the Canny operator and the Hough Transform for edge detection, lane boundaries can be highlighted effectively. Color-based detection using the HSV color space allows the system to adapt to variations in road surface and illumination, enhancing visibility and detection accuracy under diverse conditions.

- **Real-Time Processing:**

Optimized for high-quality lane detection and tracking in real-time, the algorithm is designed to minimize latency, enabling seamless execution in practical applications.

- **Indian Road Adaptation:**

The system is customized to handle challenges specific to Indian roads, such as faded, inconsistent, or partially coked lane markings; clutter on the roads; and variable weather conditions including poor lighting and rain.

- **Kalman Filter and Exponential Moving Average (EMA):**

A Kalman Filter is used to predict lane offsets accurately, even in noisy or cluttered environments, while the EMA smooths lane boundary estimations, contributing to system robustness and providing a stable lane-tracking experience.

CHAPTER-2

LITERATURE SURVEY

Subramani, R., Suresh, K., Cecil, D., & Vijayalakshmi, C. [1] This research paper primarily focuses on lane detection under different conditions which also includes different time scenarios like day and night due which the shadows created by the sun light and street lights could be a problem for detecting the lanes. The challenges faced in this study are poor lighting, different weather conditions and roads conditions. To solve these challenges they used preprocessing techniques and Kalman filter which have finally improved the overall accuracy of the project. The preprocessing techniques used in this paper are Gamma Correction, Region of interest, bilateral filtering, HSV filtering, Canny Edge detection and hough transform. R lane detection process clustering is used in this paper. Density Based Spatial Clustering of application with Noise is used for group relevant lanes. But the Limation of this paper is it doesn't work for curved lanes.

Kreucher, C., Lakshmanan, S., & Kluge, K. (1998, October) [2] This paper basically developed a system that gives an alert or intimate the driver about the lane departures with the help of LOIS algorithm for detecting lanes and Kalman filter for prediction of lanes. The Primary goal or Objective of this paper is to provide a strong driving support for the driving in particular situations like drowsiness or when driver is out of focus. The most prominent algorithm used in this paper is Lois algorithm. Lois stands for Likelihood of Image Shape. This algorithm works perfect even in conditions like shadows and varying lighting sources. At the starting of predictions the algorithm works weaker but with subsequent tracking, the algorithm uses information from previous frames which results in better accuracy and performance. This system created an intimation for the driver when the predicted position lies within the predefined threshold.

Dorj, B., Hossain, S., & Lee, D. J. (2020) [3] This Paper has implemented a system that detects curved lanes which are critical for autonomous or self-driving vehicles, specially on roads which have significant turns. It combines Kalman filter with geometric models which resulted to detect and track curved lanes with better accuracy. The primary objective of this paper is to improve the safety by predicting accurate curve road turns. The faced challenges are uncertainties in the image data like noise. They used Otsu's thresholding method and top-view image transformation for lane detection. In case of

straight line hough transform is used and in case of curve lanes, it uses Kalman filter with integration of further 2 models: parabolic model-used to fit the curve with parabolic equation and Circular model- used to fit the curve with circle equation with estimating its center and radius. This work mainly helps in adjusting the steering angle based upon the road's curvature.

Samyak Shah, A. J. (2022) [4] The paper presents a real time road lane detection system boundaries, and warns drivers of unintended lane departures and can be used for autonomous vehicle technologies. The proposed solution in the paper refines the current methods like edge detection, region of interest (ROI) segmentation, and hough transform for better accuracy and efficiency. It uses image processing tools like grayscale conversion, gaussian blur for noise reduction, and hough transform for lane line detection. The system differentiates the lane markings through color masks and focuses on a defined ROI to minimize irrelevant edge detections. The Hardware integration in this paper includes a Raspberry Pi Zero with a camera to process images and deliver real time audio warnings. It demonstrates high accuracy, i.e., over 96% even in non- optimal weather conditions, improving the road safety by preventing lane departure related accidents. The future scope includes improving curved lane detection, optimizing performance in extreme weather and incorporating databases for hazard recognition.

Rajakumar.Ra, I. P. (2021) [5] The paper presents a real time road lane detection system which can be used for reducing traffic accidents caused by not attentive driving. The system detects lane boundaries, and warns drivers of unintended lane departures and can be used for autonomous vehicle technologies. The proposed solution in the paper refines the current methods like edge detection, region of interest (ROI) segmentation, and hough transform for better accuracy and efficiency. It uses image processing tools like grayscale conversion, gaussian blur for noise reduction, and hough transform for lane line detection. The system differentiates the lane markings through color masks and focuses on a defined ROI to minimize irrelevant edge detections. The Hardware integration in this paper includes a Raspberry Pi Zero with a camera to process images and deliver real time audio warnings. It demonstrates high accuracy i.e over 96% even in non optimal weather conditions, improving the road safety by preventing lane departure related accidents. The future scope includes improving curved lane detection, optimizing performance in extreme weather and incorporating databases for hazard recognition.

Shipra Singh, S. M. (2024, May) [6] The paper titled presents a system for curved lane detection, which is crucial for vehicles and advanced driver systems (ADAS), to enhance road safety and navigation. In the paper the methodology includes combining traditional image processing with advanced algorithms, overcoming challenges such as inconsistent lane markings, lighting variations, and complex road geometries. Images are preprocessed through grayscale conversion, noise reduction, thresholding, perspective transformation for bird's eye view, and Sobel edge detection to highlight lane edges. Here, the detection process uses a sliding window technique to travel lanes and a quadratic curve fitting algorithm to represent lane curvature, which calculates the radius of curvature for precise results. The testing gives high accuracy, with robust performance across different conditions, reducing errors like false positives and lane deviations. The areas are improved as the system adapts to various environments like urban roads and mountainous terrains.

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lane markings, lighting variations, and complex road geometries. Images are preprocessed through grayscale conversion, noise reduction, thresholding, perspective transformation for bird's eye view, and Sobel edge detection to highlight lane edges. Here, the detection process uses a sliding window technique to travel lanes and a quadratic curve fitting algorithm to represent lane curvature, which calculates the radius of curvature for precise results. The testing gives high accuracy, with robust performance across different conditions, reducing errors like false positives and lane deviations. The areas are improved as the system adapts to various environments like urban roads and mountainous terrains.

The author mentioned future improvements include handling extreme weather conditions, enabling vehicle-to-vehicle communication for collective road awareness, and extending capabilities to detect pedestrian lanes and crosswalks.

Kaur, G., & Kumar, D. (2015) [7] This paper provides a brief outline of various lane detection methods which are utilized in intelligent transportation systems to improve road safety and reduce accidents. The main objective of this paper says that lane detection plays an important role in self- driving cars. This paper emphasis mainly about improving lane detection algorithms under various conditions. The existing algorithms works perfect for ideal conditions but struggles with real-world scenarios. The algorithms used in this paper is hough transform canny edge detection and bilateral filtering. Basically, this paper proposes a mixture of filtering and detection techniques to overcome the challenges.

Goel, A. (2014) [8] This paper performs different methodologies for detection of lanes in intelligent transportation systems which aims to overall improving of road safety by preventing lane deviations. Algorithms used are hough transform and canny edge detection which are effective under ideal roads but faces difficulties in real world scenarios. The advanced approaches integrates multiple algorithms with geometric models offer better accuracy. This paper identifies gaps in existing systems particularly in cases the algorithm prefers high quality images for accurate results and inability to handle extreme conditions and suggests future improvements on the algorithms for different roads and whether conditions.

Jingwei Cao, C. S. (2019) [9] The paper presents a lane detection algorithm for intelligent

vehicles that are operated in complex and critical environments. The algorithm overcomes the limitations of traditional and deep learning methods by achieving high accuracy and real time performance. It starts by correcting image distortions using camera calibration and inverse perspective transformation to provide an aerial view of the lane or view from above. The edge detection is enhanced by combining the Sobel operator and HSL color space thresholding. Lane lines are fitted using RANSAC algorithm which is based on a third order B-spline curve model, enabling strong detection even in challenging conditions. Experimental validation on road driving videos and the Tusimple dataset showed an average detection accuracy of 98.49% and 98.42%, with average processing times of approximately 21.5 ms per frame. The algorithm outclassed traditional methods and some deep-learning approaches, demonstrating excellent anti-interference, adaptability, and efficiency. It was tested under various conditions such as highways, mountain roads, and tunnels, maintaining reliable performance despite challenges like lighting, and road curvature. The study concludes that the proposed algorithm significantly enhances vehicles' safety and technical capabilities of vehicles. The future work includes aiming to improve inclusiveness and error detection to further optimize the algorithm's overall performance.

Arindol Das, [S. (2024) [10] The paper presents a lane and curve detection system using deep learning to improve safety and for reducing accidents. It incorporated traditional computer vision techniques with convolutional neural networks (CNN) for robust lane detection, particularly for curved lanes. The process starts with data preprocessing which involves gradient calculation using Sobel and Laplacian operators, Gaussian smoothing for noise reduction, and Canny edge detection for identifying lane boundaries. After preprocessing the detected lane boundaries are mapped onto original road images for visualization. The CNN model, a Fully Convolutional Network (FCN), is used to create a binary segmentation map for lane pixels. The model undergoes training with a different types of annotated datasets, and its performance is fine-tuned using various hyperparameter optimization techniques. The approach combines the accuracy of deep learning with the speed and simplicity of traditional methods. The testing on different things demonstrates its capability to adapt to various road conditions, achieving precise detection of lane markings and curvature. In Future work, further model optimizations to enhance performance and expand usability.

CHAPTER-3

DESIGN

A lane detection system for Indian roads typically involves following steps:

- **Data Preprocessing:** Frames from video input are processed, making them clear and uniform by resizing, normalization, improving the quality (contrast adjustment), and reducing noise. For Indian roads, the quality of images declines due to clutter and environmental conditions.
- **Model Architecture:** It uses a hybrid approach on traditional image processing Canny edge detection, Hough Transform- as well as HSV color-based segmentation towards the lane-marking detection. Kalman filter predicts offsets with high accuracy and EMA smooths the lane boundaries for quick tracking.
- **Evaluation:** Lane Detection Consistency, Offset Stability, FPS evaluate its reliability in detecting lanes on cluttered Indian roads.

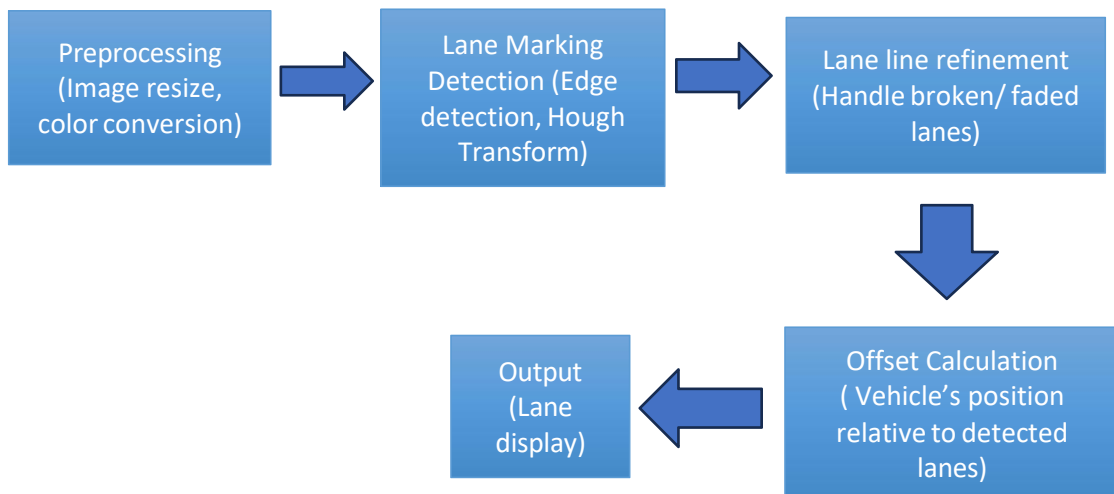


Fig 3.1 Steps in Lane Detection System (Straight)

3.1 REQUIREMENT SPECIFICATION

3.1.1 SOFTWARE REQUIREMENTS:

- **Programming Language:** Python (preferred), or any language that supports machine learning libraries (TensorFlow, OpenCV).
- **Image Processing Libraries:** OpenCV for resizing, edge detection, and color space transforms.
- **Math Library:** Using NumPy for the numerical operations involved in changing

images and filter implementations.

- Optional: Visualization libraries like Matplotlib for plotting results of detections and metrics performance.

3.1.2 HARDWARE REQUIREMENTS:

- Processor (CPU): Multi-core processor (i5 or equivalent) for smooth operation and real-time processing.
- Memory (RAM): Minimum 8 GB, recommended; more video file processing performance enhances with more RAM.
- Storage: Storage of the processed data.
- Graphics Processing Unit: Though this component is optional, having a GPU, such as NVIDIA GTX or AMD Radeon, will significantly speed up real-time processing, especially when considering deep learning.

3.2 UML DIAGRAM:

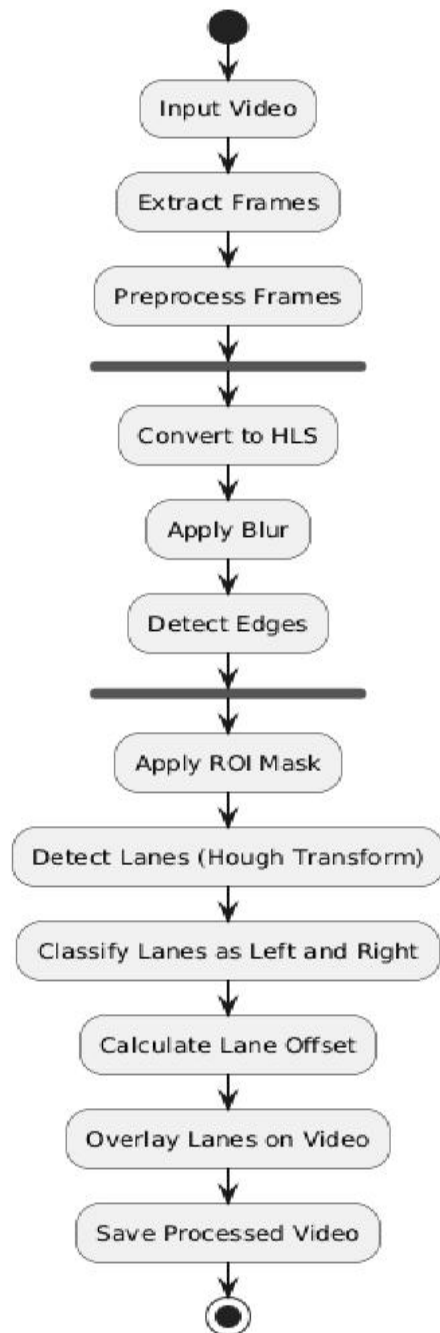


Fig 3.3 Activity Diagram

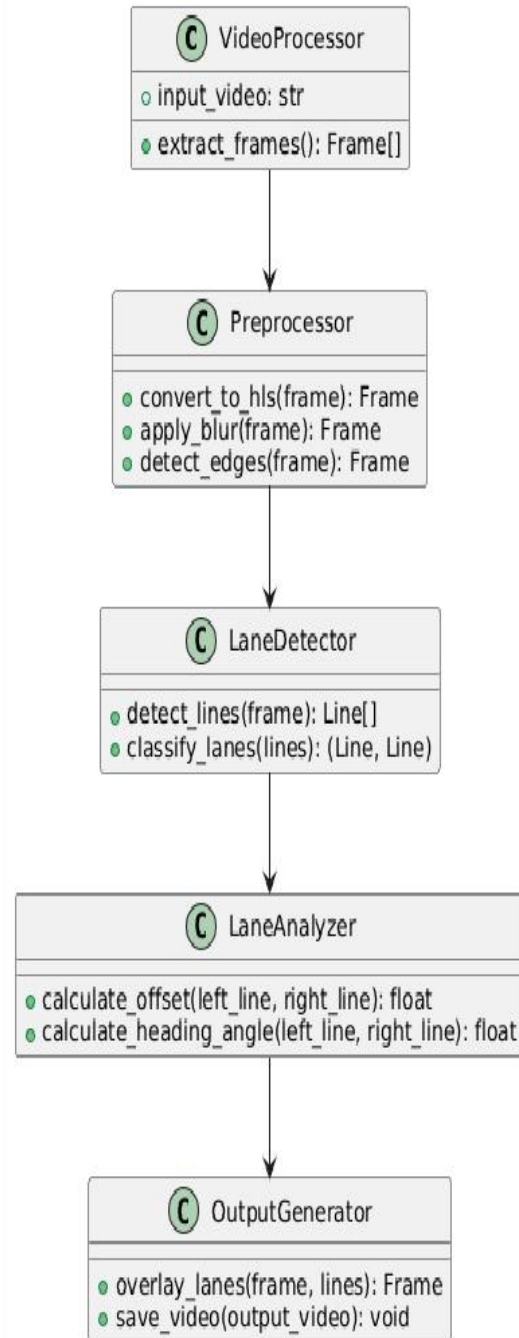


Fig 3.4 Class Diagram

CHAPTER-4

IMPLEMENTATION

The system involves following key implementation steps:

Preprocessing:

Frames from input video are processed to enhance the observability of lane markings

- **Resizing:** Each image gets resized for uniformity and speedy processing.
- **Contrast Enhancement:** Images are made brighter for enhanced readability, particularly regarding whether faded or obscured lane marking. It is done by converting video frames into HLS color space. A binary mask is applied to isolate white areas on the frame.
- **Noise Reduction:** After applying binary mask, Gaussian blur is applied to reduce noise levels and smoothen the image and, therefore, making lane markings clearer.

Edge Detection:

The system, then employs algorithms for edge detection, i.e., Canny edge detector to aid in the search for edges that might be helpful to detect lane boundaries. The line found is then considered a straight lane line from an edge-detected image using the Hough Transform function.



Fig 4.1 Input image



Fig 4.2 HLS color space



Fig 4.3 Mask white and Gaussian blur



Fig 4.4 Canny edge detection

Filtering and smoothing:

The Kalman Filter is utilized by the system to make the lane detection stable and compensate for whatever type of noise or abrupt shift in the lane. This filter uses past frame data to predict the future positions of lanes, hence making the detection more accurate and reducing the effects of noise or sudden changes in lane positions.

Real-time lane tracking is done using an EMA. It stabilizes the lane boundaries due to the smoothing of the detected lane positions through time, which amplifies the capability of the system in continuous tracking of lanes across frames.

Evaluation:

The parameters- Lane Width, Heading Angle and Offset, for every 10 frames are taken to evaluate the model performance.

- **Correlation Analysis:** The parameters Offset and Heading Angle show a strong negative correlation (-0.9) indicating the inverse relationship, which could be useful for appropriate trajectory adjustments. Whereas, the parameters Offset and Lane Width have moderate correlation (0.43) indicating some dependency.

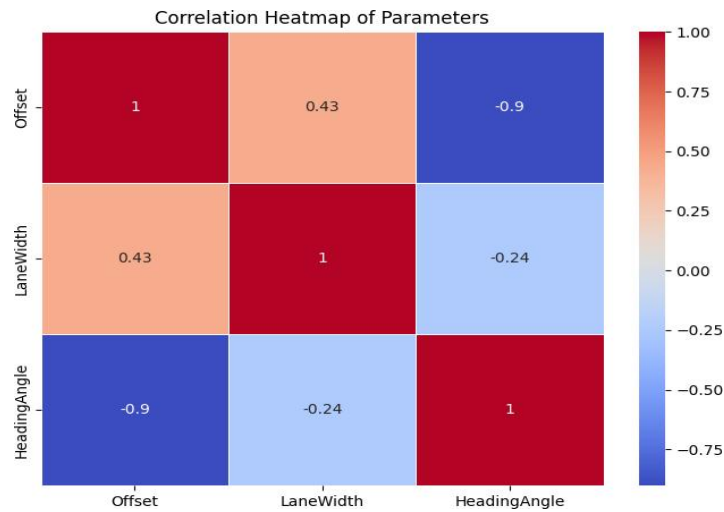


Fig 4.5 Correlation Heatmap

- **Scatter plot Interpretation:** Below is the scatter plot of Offset vs Lane Width which displays the variability in the parameters. These variations indicate road curvature, lane fading or varying lighting conditions.

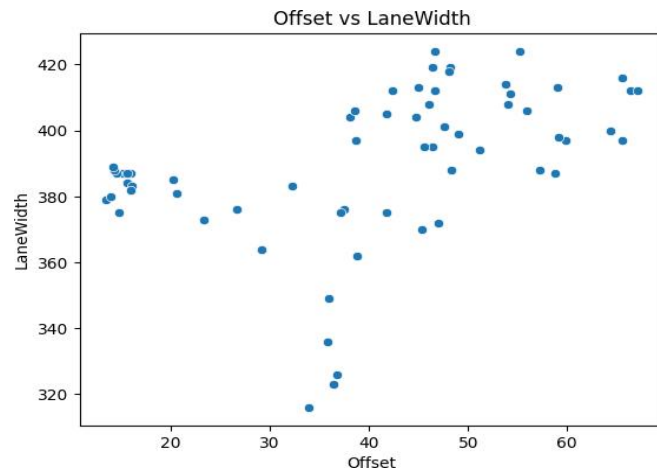


Fig 4.6 Scatter Plot

- **Behavior of Parameters:** Below is the time series graph plotted to monitor the temporal changes in the parameters and the effect of the optimizations used in stabilizing the measurements.

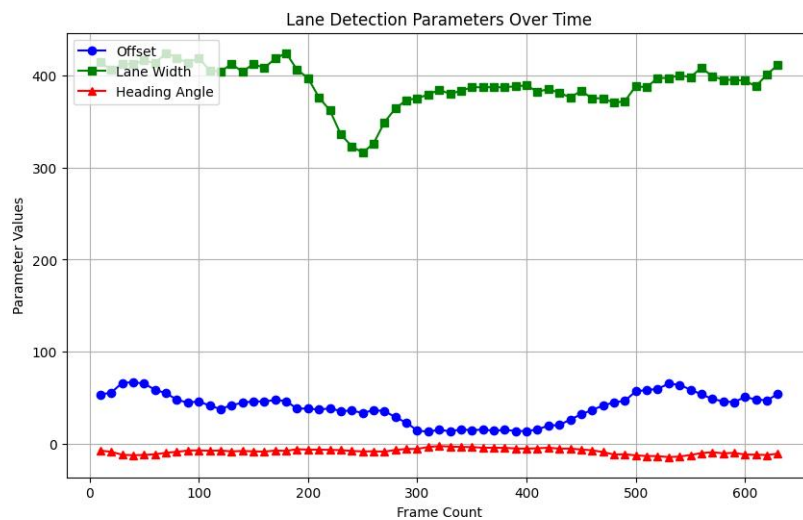


Fig 4.7 Time Series Graph

- **Lane Detection Consistency:** 99.91% consistency in detecting road lane markings under natural variations of road condition.
- **Offset Estimation Average:** 0.36 stability to predict lateral offset from lane center.
- **Lane Smoothing Track and Process:** Real-Time processing at an average of 83.32 FPS.
- **Robustness:** strong in detection of lanes of different conditions, broken or faded lines, and various road clutter types.

4.1 MODULES

Core Libraries for Image Processing:

- **OpenCV:** Used for image preprocessing (resizing, edge detection) and lane detection tasks such as Canny edge detection and Hough Transform.
- **NumPy:** Supports Performing Numerical Operations Related to Image Transformations and Matrix Computation.

Data Handling and Augmentation:

- **NumPy:** To manipulate and process the road image dataset.
- **Augmentation libraries:** To mimic diverse conditions on the road, robustly improving the model.

Data Visualization:

- **OpenCV (Real-time Visualization):** This is used to superimpose detected lanes over the video frames during real-time processing.

Lane Detection and Tracking Algorithm:

- **Kalman Filter:** Determines future lane positions; thus, no positional inaccuracies.
- **Exponential Moving Average (EMA):** This helps smooth the lane boundary and ensures stable and consistent lane tracking.

4.2 OVERVIEW TECHNOLOGY:

The Challenge:

Lane detection in Indian roads will pose a very challenging task as the road markings vary and the traffic density is heavy, etc. Thus, all these issues do cause problems in detecting lanes consistently in real-world or unstructured environments.

Why Hybrid Models Excel:

This kind of problem highly suited to a hybrid approach combining traditional image processing-such as edge detection and Hough Transform-with some real-time tracking algorithms like Kalman Filter and EMA. The proposed method is very efficient in the detection of lane boundaries, even with road noises and variable lighting, while ensuring real-time processing.

How the Model Works:

- **Input:** The system takes in road images or video frames.
- **Edge detection:** Canny edge detection and the Hough Transform are used in algorithms for detecting lane boundaries using techniques in image processing. Color- based segmentation is also used to outline lane markings under various lighting conditions.
- **Optimization:** The Kalman Filter serves to forecast lane positions across temporal intervals, whereas the Exponential Moving Average (EMA) facilitates the maintenance of smooth lane tracking, despite varying environmental conditions.
- **Lane Detection:** Image processing combined with tracking gives the accurate lane detection as well as real time position estimations of the lanes.

Benefits:

- **High Accuracy:** The hybrid model can obtain very high accuracy in lane detection with difficult roadway conditions such as debris, worn lane markings, and inadequate illumination. The system can process real-time video frames, so it is very feasible to use in autonomous vehicle applications.
- **Robustness:** The model withstands road noise, changes in traffic conditions, and styles of lane markings on Indian roads effectively.

Below tables shows the results obtained by processing the same input for Kalman optimization, EMA and hybrid approach.

<i>Optimization</i>	<i>Lane Detection Consistency</i>	<i>Offset stability</i>	<i>FPS</i>
<i>Kalman Filter</i>	99.52%	16.69 px	69.68
<i>Exponential Moving Average (EMA)</i>	99.52%	39.74 px	81.88
<i>Hybrid Offset (Kalman + EMA)</i>	99.88%	0.53 px	79.94

Table 4.1 Comparison of different methods

The hybrid method outperforms both Kalman and EMA in lane detection consistency and offset stability, while maintain a high FPS. This combination could be potential approach for real-time lane detection with good accuracy and performance speed.

Materials for Further Reading:

- Hybrid models for lane detection methodologies can provide interesting insights from scholarly articles on what may come next.
- Tutorials on the usage of OpenCV and some Python libraries like NumPy and Matplotlib can implement and visualize lane detection systems.

4.3. OPTIMIZATION TECHNIQUES FOR LANE DETECTION:

Optimization is a crucial step in ensuring that a lane detection system operated with high accuracy, stability and responsiveness. In real-world driving scenarios, lane markings can be faint, broken or obstructed, requiring a system that can track lanes smoothly without abrupt shifts. To achieve this, Exponential Moving Average (EMA) smoothing was used to stabilize lane tracking, and Bayesian Optimization was applied to fine-tuned the EMA parameters for the best performance.

IMPORTANCE OF OPTIMIZATION IN LANE DETECTION:

A lane detection system must perform well under varied conditions such as:

Sharp curves and lane merges, where abrupt changes in lane positioning can cause instability.

1. Faded or broken lane markings, which make detection inconsistent.
2. Shadows, glare and low-light conditions where lane contrast is reduced.
3. High-speed driving scenarios, where rapid lane tracking adjustments are necessary.
4. Without proper optimization, a lane detection system might either:
5. Respond too slowly, causing delayed lane updates.
6. React too aggressively, leading to unstable lane tracking.

By fine-tuning the EMA smoothing factor, we ensured a balance between smooth tracking and fast response times, making the system both stable and adaptable.

WHY EMA SMOOTHING?

Exponential Moving Average (EMA) is a technique used to smooth lane position estimations across frames. Instead of reacting to each frame's detection independently,

EMA assigns more weight to past lane positions, ensuring a gradual and stable transitions between frames.

EFFECT OF ALPHA ON LANE STABILITY:

- Low Alpha (0.2 - 0.5) :
 - Produces smoother lane tracking by heavily averaging past detection's.
 - Can be slow to react to sudden lane changes (sharp curves)
- High Alpha (0.8 to 1.0):
 - Fast response to lane changes but introduces instability in tracking.
 - More sensitive to noisy detection's, leading to jittering.

BAYESIAN OPTIMIZATION FOR FINE-TUNING EMA

Bayesian Optimization is a technique used to find the best hyper-parameters by systematically evaluating different values and improving predictions based on prior results. Instead of manually testing different alpha values, Bayesian Optimization selects and refines parameters automatically reducing trial-and-error efforts.

HOW IT WORKS?

STEP 1: DEFINE THE OBJECTIVE FUNCTION:

The system evaluated lane tracking performance based on smoothness stability and real-time adaptability.

STEP 2: BAYESIAN OPTIMIZATION BEGINS:

The algorithm starts by testing a few initial alpha values and measuring how well they perform.

STEP 3: ITERATIVE LEARNING:

Using past results, Bayesian Optimization predicts better alpha values for the next iterations, converging towards the optimal value.

STEP 4: CONVERGENCE TO BEST ALPHA:

After several iterations, the system identifies the best alpha (0.64) which provides the best balance between lane stability and responsiveness.

GRAPHICAL ANALYSIS:

GRAPH1: BAYESIAN OPTIMIZATION CONVERGENCE:

This graph shows how Bayesian Optimization progressively improves the lane tracking stability. Initially, different alpha values cause fluctuations, but the system gradually converges to the optimal setting, ensuring smooth tracking.

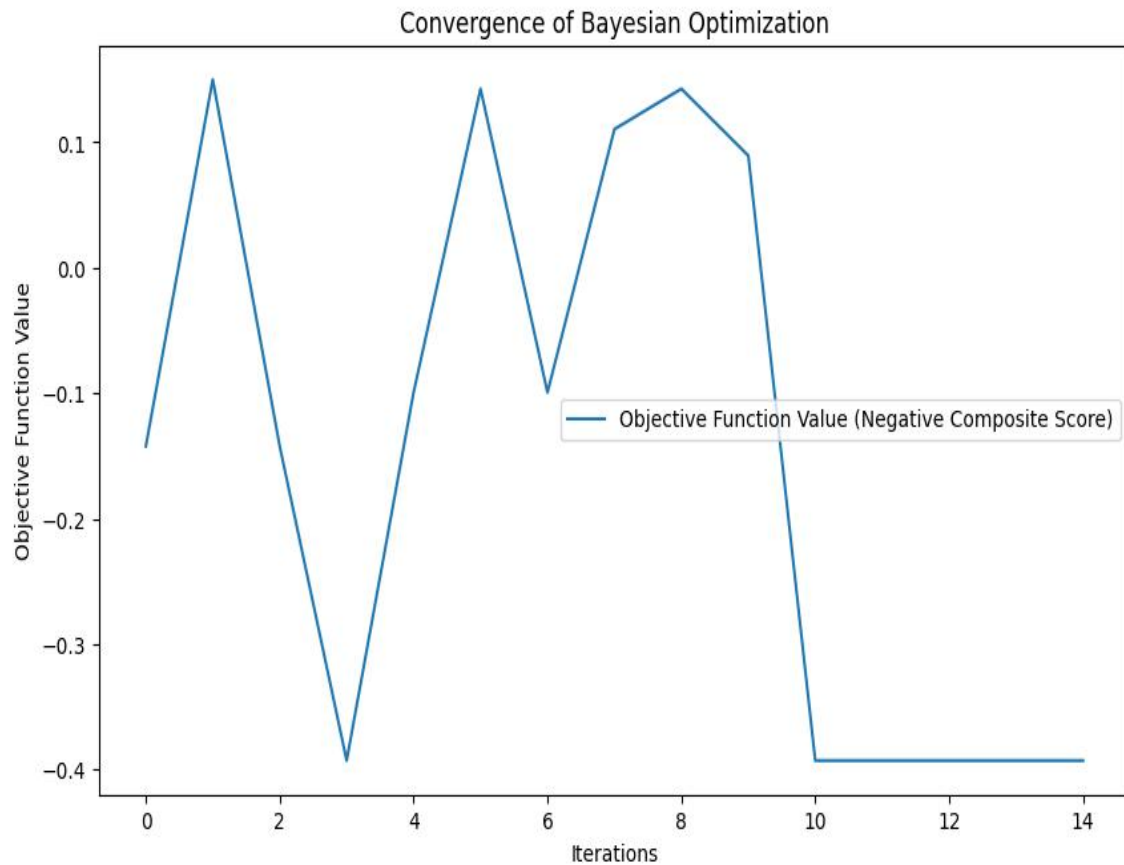


Fig 4.8 Convergence of Bayesian Optimization

GRAPH 2: EFFECT OF ALPHA ON LANE OFFSET:

This graph highlights how different alpha values affect lane tracking:

Low alpha results in smooth but delayed lane updates.

High alpha results in fast but unstable tracking.

Optimal alpha (0.64) balances both ensuring stability without lag.

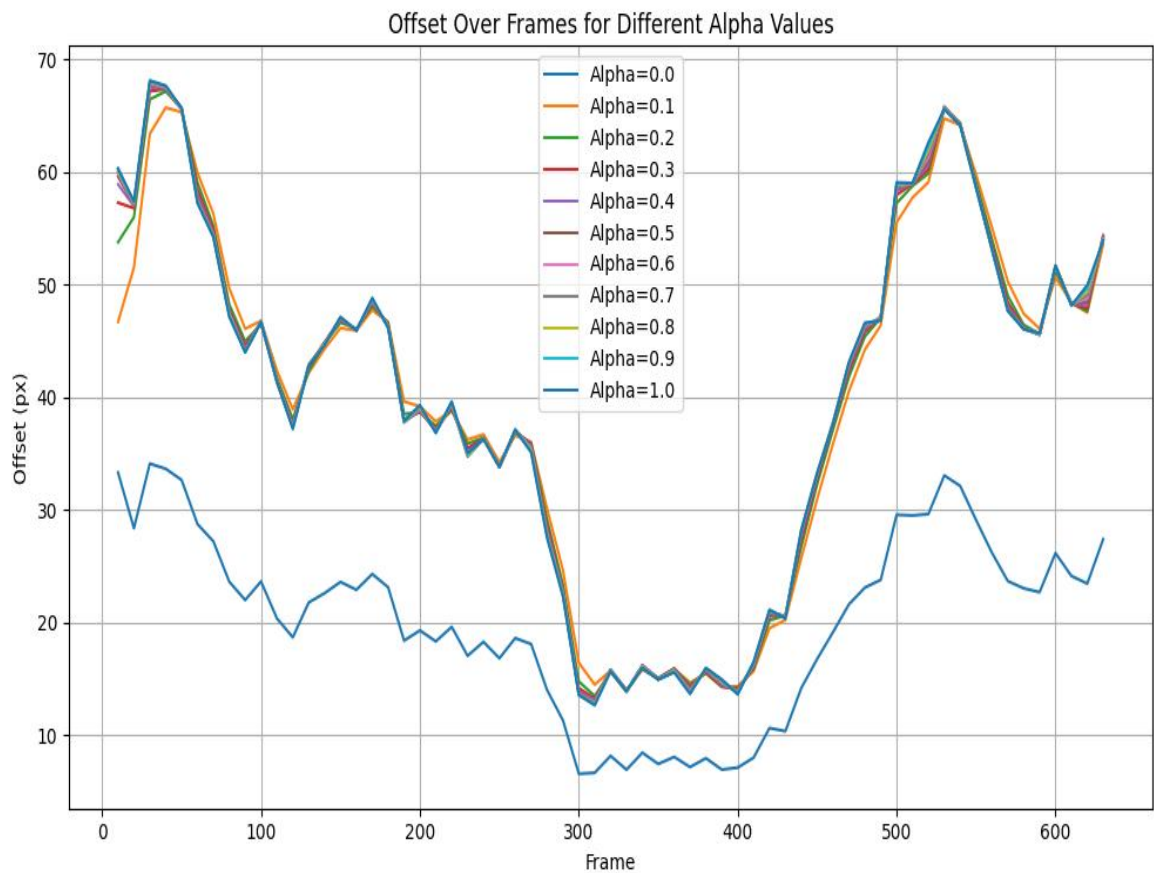


Fig 4.9 Offset over frames for different alpha values

4.4 POST PROCESSING IN LANE DETECTION:

Post-processing is a crucial step in lane detection systems that refines the outputs from the deep learning model ensuring stability, accuracy and noise reduction. The raw lane predictions generated by the U Net segmentation model may contain gaps or unstable lane positioning, making it essential to apply post-processing techniques before final visualization.

4.5 DATASET PREPARATION:

A custom data-set was created by extracting video frames from traditional lane detection outputs. Each frame was re-sized to 224 * 224 pixels to ensure uniformity in training. The dataset was then labeled to create a binary lane masks, where masked as white (1) and the background as black (0).

STEPS IN DATASET PREPARATION:

1. FRAME EXTRACTION: Captures images from lane detection videos.
2. RESIZING: Standardize all images to 224 * 224 pixel for consistency .
3. LANE MASKING:Used HSV color threshold to isolate lane markings.
4. DATA AUGMENTAON: Applied flipping , rotation , brightness changes to improve model robustness.

This dataset was then used to train a U Net model for lane segmentation.

4.6 DEEP LEARNING MODEL:

The U Net architecture was chosen for lane detection because of its ability to perform pixel-wise segmentation while preserving spatial details. The model consists of two main components :

ENCODER: Extracts lane-related features using convolutional layers.

DECODER: Reconstructs lane masks from encoded features using upsampling layers.

6.6.1 TRAINING PROCESS:

INPUT: 224 * 224 images with normalized pixel values.

OUTPUT: Binary mask indicating lane positions.

LOSS FUNCTION: BCEWithLogitsLoss for pixel classification.

OPTIMIZER: Adam for efficient weight updates.

HARDWARE: Trained using GPU acceleration for faster convergence.

Once trained the model was integrated into the lane detection pipeline, where post-

processing was applied to improve real-world performance.

4.7 POST-PROCESSING TECHNIQUES FOR LANE STABILITY:

4.7.1 MORPHOLOGICAL FILTERING:

After obtaining the lane mask from the U Net model, morphological operations were applied to improve lane continuity. These operations helped in:

- Removing small noise pixel that were misclassified as lanes.
- Connecting broken lane markings by filling small gaps.

4.7.2 LANE OFFSET CALCULATION:

To determine the vehicle's position relative to the lane, the lane offset was calculated using:

$$\text{Lane offset} = \text{lane center} - \text{vehicle center}$$

Lane center: Midpoint between detected left and right lanes.

Vehicle center: center of the camera frame.

This calculation helps in assessing lane deviation, which is crucial for lane keeping systems.

4.7.3 STABILIZING LANE TRACKING WITH KALMANN FILTERING AND EMA:

Even after lane detection, slight variations across frames can cause instability. To smooth the tracking a Kalman Filter + EMA hybrid approach was implemented.

Kalman Filter: Predicts lane positions in future frames, ensuring stable tracking even during temporary occlusions.

Exponential Moving Average (EMA): Reduces flickering by averaging lane positions over time, improving tracking stability.

4.8 PERFORMANCE EVALUATION:

- Smoother lane transitions due to Kalman + EMA filtering.
- Significant noise reduction through morphological processing.
- Improved lane continuity, even when markings were faded.
- Accurate vehicle positioning using lane offset calculations.

Graphical results confirmed that post-processing significantly enhanced lane stability and reduced detection errors.

4.8 INTEGRATION OF TRADITIONAL AND DEEP LEARNING APPROACHES:

To achieve both speed and accuracy, a hybrid approach was implemented:

- Traditional Methods (Edge Detection & Hough Transform) provided fast initial lane estimates.
- Deep Learning (U Net) refined the detection for better accuracy in complex conditions.
- Post-Processing (Kalman Filter & EMA) stabilized lane tracking over time.

This integration ensured that the system was both computationally efficient and highly accurate.

CHAPTER-5

TESTING

Test Cases for Evaluation

1. **Obtaining Video Streams:** Capabilities test the system's capability to successfully acquire video streams from uploaded video recordings. Ensure that the video feed taken matches the standards necessitated for lane detection and tracking.
2. **Pre processing Check:** Verify correct application of pre processing operations including re-sizing and edge detection on every video frame. Confirm whether color space transformation (HSV) and noise reduction processes such as Gaussian blur are applied to enhance the lane detection model accuracy.
3. **Lane Detection Using Edges and Color-Based Techniques:** Assess the robustness of the system in detecting lanes in various lighting, meteorological, and roadway conditions. The accuracy of lane localization should be verified: Lane markings in video frames must be correctly represented.
4. **DEEP LEARNING-BASED LANE DETECTION EVALUATION:** Evaluate the UNet model's lane detection accuracy using IoU. Verify lane consistency across frames, ensuring robustness against occlusions, faded lines and false positives.
5. **Offset Estimation with Kalman Filter and EMA:** Evaluate the effectiveness of the Kalman Filter in accurately estimating the car's offset from the center of the lane for contiguous frames. The effectiveness of the Exponential Moving Average in the reduction of lane boundaries and in maintaining an offset measurement.
6. **Lane Monitoring and Continuity:** The identified lane should also have been monitored without meaningful disruptions at time t consecutive frames. Test the tracking robustness for faded or broken lanes, or when other vehicles partially obscure the lanes.
7. **Post processing Verification:** Confirm the correctness of lane boundary predictions and make sure that only the items related to lane markings should not appear in the output. Confirm that your final product is good, and ready for the video where it will be demonstrated indicating continuous lane tracking for viewing.

CHAPTER-6

RESULTS

Comparison of results of previous models to our model under different conditions:



Fig 6.1 Condition: Broken Lanes



Fig 6.2 Condition: Shadows



Fig 6.3 Condition: Expansion Joints



Fig 6.4 Output on Indian Roads

CHAPTER-7

CONCLUSION

The advanced lane detection system successfully designed has been able to successfully perform real-time identification of lanes and the tracking of a vehicle. The achievement was specially impressive considering the generally adverse road conditions, such as bad lane markings, the incidence of shadows, and the presence of expansion joints on the road. The proposed model, which provides a combination of conventional image processing methods with hybrid filtering techniques, is eligible to clearly detect the boundaries of lanes. In addition to this, it also makes accurate calculations of offsets so that the vehicle is exactly in the center of its allocated lane at any point of time in its journey.

This particular approach presents an exciting and motivational solution towards enhancing road safety and simultaneously facilitating the design and incorporation of autonomous driving technology. It is significant because vast and challenging road conditions are often found on Indian roads where general road environments can be extreme in many areas. Results obtained from this experiment show high detection and excellent real-time processing capabilities. These are very important results because they provide a basis from which further improvements can be engineered, especially targeted towards achieving enhanced accuracy in lane detection and overall stability in the systems. Some of the possible opportunities for interventions could be deep learning techniques that maximize the adaptability of such systems. The result would significantly contribute to the overall reliability of these intelligent driving technologies, ensuring better resilience in different conditions under which driving becomes operational.

CHAPTER-8

FUTURE SCOPE

The development of a Lane detection system utilizing OpenCV and other technologies offers numerous opportunities for future research and advancement:

- **Deep Learning Improved Lane Detection:** Techniques such as CNNs, which evolved into Transformer-based architectures in the newest deep learning architecture, could significantly improve the accuracy level of lane detection and allow following curvy roads, complex structures, and lanes that are poorly marked.
- **Improving Adaptability to Varying Road Surfaces:** This system would be supposed to improve on the functionality of lane detection responding to adverse conditions such as ill lit roads, extreme weather, heavy traffic, and others.
- **Real-time Performance Optimization:** It can be further optimized to cut down processing time and resource consumption to achieve flawless real-time lane detection for applications in autonomous driving and ADAS.
- **Dynamic Lane Change and Lane Obstacle Detection:** Developing the model for dynamic lane changes such as lane splits, exits, or merging lanes and detecting lane obstacles would make the system highly versatile and straightforward vehicle navigation.
- **ITS Integration:** V2I communication integration would therefore allow connectivity with the smart city infrastructure and, thereby, enable coordinated traffic management, adaptive routing, and automated response to lane usage and flow in traffic.
- **Collection of Data and Ongoing Learning:** Employ mechanisms of ongoing learning where the model is refreshed at regular intervals by new data taken from multiple different environments. This way, the model remains fresh and thus effective in different types of roads and geographies.
- **Real-time Feedback for Driver Assistance:** Even with real-time feedback, lane deviation or potential lane departure warnings can easily support driver safety and enhance the current ADAS technologies provided in commercial vehicles. Exploring these future directions might make your lane detection project even significantly contribute toward the development of road safety.

CHAPTER-9

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[**YOUTUBE LINK**](#)

[**LINKEDIN LINK**](#)